SIMATS ENGINEERING

ASSESSMENT TOOL 3

CLASS TEST

Course Name: Data Warehousing and Data Mining

Course Code: CSA16

Course Outcome: CO2 – Apply suitable Data Pre-processing techniques like Data Cleaning, Data Integration, Data Transformation and Data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 20%   **Total Questions:** 10   **Total Marks:** 20

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PROBLEM STATEMENT

Assessment Tool 3 – Class Test

Q1. Data Mining and KDD Process

Define **Data Mining** and list any two basic data mining tasks. Explain the complete steps involved in the **Knowledge Discovery in Databases (KDD)** process and justify the importance of **data preprocessing** within it. **[L2, L4]**

KDD Flow: Data Sources → Data Selection → Data Cleaning → Data Integration → Data Transformation → Data Mining → Pattern Evaluation → Knowledge Presentation → Useful Knowledge.

Q2. Data Cleaning

What is **data cleaning**? Mention two common data quality issues. Explain methods used to handle **missing values** and **noisy data** with suitable examples. **[L1, L2, L3]**

Discuss missing data, noisy data, inconsistent data, duplicate records, mean/median/mode imputation, regression-based imputation, k-NN imputation, binning, regression, clustering, and outlier treatment.

Q3. Data Integration

Define **data integration** and list two challenges associated with it. Explain how **redundancy** and **inconsistency** are handled during data integration. **[L2, L4]**

Discuss schema integration, entity identification, redundancy detection, naming conflicts, format conflicts, duplicate attributes, transformation, and standardization.

Q4. Covariance Analysis

What is **covariance analysis**? List the two covariance formulas. **[L2, L3]**

Population: $Cov(X,Y) = \Sigma[(x_i – x\blacksquare)(y_i – \blacksquare)] / N$

Sample: $Cov(X,Y) = \Sigma[(x_i – x\blacksquare)(y_i – \blacksquare)] / (N – 1)$. Explain positive, negative, and zero covariance.

Q5. Covariance and Stock Prices

Suppose two stocks **A** and **B** have the following values during one week. If the stocks are affected by the same industry trends, determine whether their prices are likely to rise together, fall together, or move independently. Use covariance analysis to support the answer.

Stock A	Stock B
2	5
3	8
5	10
4	11
6	14

Required analysis: Show the relevant calculations, tables, interpretation, and final conclusion clearly.

Q6. Architecture of a Typical Data Mining System

Explain the architecture of a typical **Data Mining System** with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. **[L2, L4]**

Architecture: Data Sources → Database/Data Warehouse → Data Cleaning → Data Integration → Data Selection → Data Transformation → Data Mining → Pattern Evaluation → Knowledge Presentation. Explain each component.

Q7. Covariance Calculation

Find the covariance for the following dataset:

x = {4, 5, 6, 7, 9}
y = {8, 3, 7, 5, 6}

Calculate number of observations, means, deviations, products of deviations, sum of products, covariance, and interpretation. Use population covariance unless otherwise specified.

Q8. Data Pre-processing Pipeline

Describe the major steps involved in **Data Pre-processing**. Given a dataset containing missing values, outliers, and inconsistent formats, propose a systematic preprocessing pipeline and justify the sequence of steps. **[L3, L4, L5]**

Suggested pipeline: Raw Dataset → Data Understanding → Missing Value Detection → Missing Value Treatment → Duplicate Detection → Outlier Detection → Outlier Treatment → Format Standardization → Data Integration → Data Transformation → Data Reduction → Data Validation → Clean Dataset → Data Mining / Machine Learning.

Q9. Equal-Frequency Binning

A hospital is analyzing patient age data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency (equi-depth) binning**. **[L2, L4]**

Sample Dataset – Ages: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Determine the number of observations, divide into approximately equal-frequency bins, show each bin, calculate smoothing values, produce the final smoothed dataset, and discuss the effect of age 70.

Required analysis: Show the relevant calculations, tables, interpretation, and final conclusion clearly.

Q10. Data Discretization and Binning

Explain **data discretization** and its importance in Data Mining. Compare **equal-width** and **equal-frequency** binning with suitable examples, and evaluate their impact on data analysis.

Cover definition, purpose, continuous attributes, categorical representation, equal-width binning, equal-frequency binning, advantages, limitations, applications, and impact on subsequent Data Mining.

TOPIC COVERAGE

Question	Topic
Q1	Data Mining and KDD
Q2	Data Cleaning
Q3	Data Integration
Q4	Covariance Analysis
Q5	Covariance Application
Q6	Data Mining System Architecture
Q7	Covariance Calculation
Q8	Data Pre-processing Pipeline
Q9	Equal-Frequency Binning
Q10	Data Discretization

EXPECTED LEARNING OUTCOME

After completing the assessment, the student should be able to explain fundamental Data Mining concepts, describe the KDD process, identify data quality problems, apply data cleaning and integration concepts, calculate and interpret covariance, explain Data Mining system architecture, design a systematic preprocessing pipeline, apply equal-frequency binning, and compare discretization techniques.

The assessment is intentionally structured as a multi-page problem statement document so that each major question and topic can be reviewed clearly before attempting the Class Test.